

Homework 7

V.6

Problem 1. Calculate the terms through order seven of the power series expansion about $z = 0$ of the function $1/\cos(z)$.

Solution: We can write

$$\begin{aligned}\frac{1}{\cos(z)} &= \frac{1}{1 - \frac{z^2}{2!} + \frac{z^4}{4!} - \frac{z^6}{6!} + O(z^8)} \\ &= \frac{1}{1 - \left(\frac{z^2}{2!} - \frac{z^4}{4!} + \frac{z^6}{6!} + O(z^8)\right)} \\ &= 1 + \left(\frac{z^2}{2!} - \frac{z^4}{4!} + \frac{z^6}{6!} + O(z^8)\right) + \left(\frac{z^2}{2!} - \frac{z^4}{4!} + \frac{z^6}{6!} + O(z^8)\right)^2 \\ &\quad + \left(\frac{z^2}{2!} - \frac{z^4}{4!} + \frac{z^6}{6!} + O(z^8)\right)^3 + O(z^8) \\ &= 1 + \frac{1}{2}z^2 + \frac{5}{24}z^4 + \frac{61}{720}z^6 + O(z^8)\end{aligned}$$

□

Problem 3. Show that

$$\frac{e^z}{1+z} = 1 + \frac{1}{2}z^2 - \frac{1}{3}z^3 + \frac{3}{8}z^4 - \frac{11}{30}z^5 + \dots$$

Show that the general term of the power series is given by $a_n = (-1)^n \left[\frac{1}{2!} - \frac{1}{3!} + \dots + \frac{(-1)^n}{n!} \right]$, $n \geq 2$.

What is the radius of convergence of the series?

Solution: Note that

$$e^z = 1 + z + \frac{1}{2}z^2 + \frac{1}{6}z^3 + \frac{1}{24}z^4 + \frac{1}{120}z^5 + \dots$$

and

$$\frac{1}{1+x} = \frac{1}{1-(-x)} = 1 - z + z^2 - z^3 + z^4 - z^5 + \dots$$

So we have,

$$\frac{e^z}{1+z} = \left(1 + z + \frac{1}{2}z^2 + \frac{1}{6}z^3 + \frac{1}{24}z^4 + \frac{1}{120}z^5 + \dots\right)(1 - z + z^2 - z^3 + z^4 - z^5 + \dots)$$

$$= 1 + \frac{1}{2}z^2 - \frac{1}{3}z^3 + \frac{3}{8}z^4 - \frac{11}{30}z^5 + \dots$$

Note that in general we have

$$e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!} \quad \text{and} \quad \frac{1}{1+x} = \sum_{n=0}^{\infty} (-1)^n z^n.$$

In general we have,

$$\begin{aligned} a_n &= 1 \cdot (-1)^n + 1 \cdot (-1)^{n-1} + \frac{1}{2!}(-1)^{n-2} + \frac{1}{3!}(-1)^{n-3} + \dots + \frac{1}{n!}(-1)^0 \\ &= (-1)^n \left[1 - 1 + \frac{1}{2!}(-1)^{-2} + \frac{1}{3!}(-1)^{-3} + \dots + \frac{1}{n!}(-1)^{-n} \right] \\ &= (-1)^n \left[\frac{1}{2!} - \frac{1}{3!} + \dots + \frac{(-1)^n}{n!} \right]. \end{aligned}$$

Since $\frac{1}{1+z} = \sum_{n=0}^{\infty} (-1)^n z^n$ has radius of convergence of 1 and e^z has infinite radius of convergence, then we must have that $\frac{e^z}{1+z} = \sum_{n=0}^{\infty} a_n z^n$ has a radius of convergence 1. $\square$

Problem 6. Show that the coefficients of a power series “depend continuously” on the function they represent, in the following sense. If $(f_m(z))_m$ is a sequence of functions which are analytic and converge uniformly to $f(z)$ for $|z| < \rho$, and $f_{m(z)} = \sum_{k=0}^{\infty} a_{k,m} z^k$, $f(z) = \sum_{k=0}^{\infty} a_k z^k$, then for each $k \geq 0$ we have $a_{k,m} \rightarrow a_k$ as $m \rightarrow \infty$.

Proof. Let $\varepsilon > 0$, $0 < r < \rho$, and $k \geq 0$. By uniform convergence we have that there exists $M \in \mathbb{N}$ such that for all $m > M$, $z \in \mathbb{C}$ we have $|f_m(z) - f(z)| < \varepsilon r^k$. Note by Cauchy’s integral formula we have that

$$a_k = \frac{1}{2\pi i} \int_{|z|=r} \frac{f(z)}{z^{k+1}} dz \quad \text{and} \quad a_{k,m} = \frac{1}{2\pi i} \int_{|z|=r} \frac{f_m(z)}{z^{k+1}} dz.$$

Then we have

$$\begin{aligned} |a_{k,m} - a_k| &= \frac{1}{2\pi} \left| \int_{|z|=1} \frac{f_m(z) - f(z)}{z^{k+1}} dz \right| \\ &\leq \frac{1}{2\pi} \int_{|z|=1} \frac{|f_m(z) - f(z)|}{|z^{k+1}|} |dz| \\ &< \frac{1}{2\pi} \int_{|z|=1} \frac{\varepsilon r^k}{r^{k+1}} |dz| \\ &= \frac{\varepsilon}{r} \frac{1}{2\pi} 2\pi r = \varepsilon. \end{aligned}$$

So we are done. $\square$

V.7

Problem 1. Find the zeros and orders of zeros of the following:

(a) $(z^2 + 1)/(z^2 - 1)$. (c) $z^2 \sin(z)$. (e) $(\cos(z) - 1)/z$.

Solution to (a): Notice that we can write

$$\frac{z^2 + 1}{z^2 - 1} = \frac{(z - i)(z + i)}{z^2 - 1}.$$

So we clearly have that $i, -i$ are zeros. Since $\frac{1}{z^2 - 1}$ doesn't have any zeros than these are all of them.

Notice that we can write

$$\frac{z^2 + 1}{z^2 - 1} = (z - i) \frac{z + i}{z^2 - 1} \text{ and } \frac{z^2 + 1}{z^2 - 1} = (z - (-i)) \frac{z - i}{z^2 - 1}.$$

So by definition of the order of zeros, we have that both i and $-i$ are zeros of order 1. □

Solution to (c): Note that z^2 has an zero at $z = 0$ with order 2. We can also observe that $\sin(z)$ has a zero at $z = 0$ with order 1 since $(\sin(z))'|_{z=0} = \cos(z)|_{z=0} = 1 \neq 0$. So to get the order of $z = 0$ for $z^2 \sin(z)$ we simply add the orders for each function. Hence the $z = 0$ is a zero of order 3. Also observe that z^2 only has a zero at $z = 0$, but $\sin(z)$ has a zero at every integer multiple of π , i.e. $z = k\pi$, for $k \in \mathbb{Z}$. We already took care of the $k = 0$ case, so we focus on $k \in \mathbb{Z} \setminus \{0\}$. Notice, $(\sin(z))'|_{z=k\pi} = \cos(z)|_{z=k\pi} = (-1)^k \neq 0$. Hence $\sin(z)$ has zero $z = k\pi$ of order 1 for $k \in \mathbb{Z} \setminus \{0\}$. Thus $z^2 \sin(z)$ also has zero $z = k\pi$ of order 1 for $k \in \mathbb{Z} \setminus \{0\}$. □

Solution to (e): Note that $\frac{1}{z}$ has no zeros. So it suffices to find the zeros and their orders for $\cos(z) - 1$. Notice that this happens when $\cos(z) = 1$. So $z = 2\pi k$ are the zeros for $\cos(z) - 1$ where $k \in \mathbb{Z}$. However we must disclude $z = 0$ since $\frac{1}{z}$ is undefined there. So the zeros of $\frac{\cos(z) - 1}{z}$ are at $z = 2\pi k$ for $k \in \mathbb{Z} \setminus \{0\}$. Notice that $(\cos(z) - 1)' = -\sin(z)$ which is 0 at $z = 2\pi k$ for any $k \neq 0$. But, $(\cos(z) - 1)''|_{z=2\pi k} = -\cos(z)|_{z=2\pi k} = -1 \neq 0$. Hence the $z = 2\pi k$ is a zero of order 2 for every $k \in \mathbb{Z} \setminus \{0\}$. □

Problem 3. Show that the zeros $\sin(z)$ and $\tan(z)$ are all simple.

Proof. Note that the zeros of $\sin(z)$ are $z = n\pi$ for $n \in \mathbb{Z}$. They each have order 1 since $(\sin(z))' = \cos(z)$ which is not zero for any $n\pi$. So, each $z = n\pi$ is of order 1. Note, $\tan(z) = \frac{\sin(z)}{\cos(z)}$. We know $\frac{1}{\cos(z)}$ has no zeros and $\cos(z) \neq 0$ for any $n\pi$, i.e. $\frac{1}{\cos(z)}$ is defined for all the zeros of $\sin(z)$. So the same conclusions for $\sin$ also follow for $\tan$. □

Problem 6. Suppose $f(z)$ is analytic on a domain D and $z_0 \in D$. Show that if $f^{(m)}(z_0) = 0$ for $m \geq 1$, then $f(z)$ is constant on D .

Proof. Since f is analytic on D and $z_0 \in D$, then we can write it's power series centered at z_0 on some disk $D(z_0, r)$, where $r > 0$. We have

$$f(z) = \sum_{m=0}^{\infty} a_m (z - z_0)^m.$$

By Taylor's theorem we have

$$a_m = \frac{f^{(m)}(z_0)}{m!}.$$

But for all $m \geq 1$, $f^{(m)}(z_0) = 0 \implies a_m = 0$ for all $m \geq 1$. So $f(z) = f(z_0)$ for all $z \in D(z_0, r)$. We can define a function $g(z) = f(z_0)$ for all $z \in D(z_0, r)$. Then for every $z \in D(z_0, r) \subseteq D$ we have that $f(z) = g(z)$. By the Identity Theorem we have $f(z) = g(z) = f(z_0)$ for all $z \in D$. Hence f is constant on D □

Problem 9. Show that if the analytic function $f(z)$ has a zero of order N at z_0 , then $f(z) = g(z)^N$ for $g(z)$ analytic near zero and $g'(z_0) \neq 0$.

Proof. By definition of a zero of order N , we have $f(z) = (z - z_0)^N h(z)$ where h is analytic and non zero for $z = z_0$. Since h analytic and in particular continuous, then $h(z) \neq 0$ for $z \in D(z_0, r)$ for some $r > 0$. Then we have for z on such a disk we can take $(h(z))^{\frac{1}{N}} := e^{\frac{1}{N} \log(h(z))}$ since $h(z)$ non zero on a disk neighborhood of z_0 . We also have that $h(z)^{\frac{1}{N}}$ is analytic. So we have $f(z) = ((z - z_0)h(z)^{\frac{1}{N}})^N$ for $z \in D(z_0, r)$. Define $g(z) := (z - z_0)h(z)^{\frac{1}{N}}$ for $z \in D(z_0, r)$. Note that $g'(z_0) = h(z_0)^{\frac{1}{N}} + (z_0 - z_0)(\frac{1}{N})h(z_0)^{\frac{1}{N}-1}h'(z_0) = h(z_0)^{\frac{1}{N}} \neq 0$ since $h(z_0) \neq 0$. So we are done since $f(z) = g(z)^N$. □

VI.1

Problem 1. Find all possible Laurent expansions centered at 0 of the following functions:

(c) $1/((z^2 - 1)(z^2 - 4))$

Solution of (c): There are 3 regions of interest: $\{0 \leq |z| < 1\}$, $\{1 < |z| < 2\}$, $\{2 < |z| < \infty\}$. Note we can use PFD to get

$$\frac{1}{(z^2 - 1)(z^2 - 4)} = \frac{-1/3}{z^2 - 1} + \frac{1/3}{z^2 - 4}.$$

Region 1: $\{0 \leq |z| < 1\}$

We have that $|z^2| < 1$ and $|(z/2)^2| < 1$ so we get,

$$\begin{aligned} \frac{-1/3}{z^2 - 1} + \frac{1/3}{z^2 - 4} &= \frac{1/3}{1 - z^2} - \frac{1/12}{1 - (z/2)^2} \\ &= \sum_{n=0}^{\infty} \frac{1}{3} z^{2n} - \sum_{n=0}^{\infty} \frac{1}{12} \frac{1}{2^{2n}} z^{2n} \end{aligned}$$

$$= \sum_{n=0}^{\infty} \left(\frac{1}{3} - \frac{1}{12} \frac{1}{2^{2n}} \right) z^{2n}.$$

Region 2: $\{1 < |z| < 2\}$

We have that $|z| > 1 \implies \left| \frac{1}{z} \right| < 1 \implies \left| \frac{1}{z^2} \right| < 1$ and $|z| < 2 \implies \left| \frac{z}{2} \right| < 1 \implies \left| \left(\frac{z}{2} \right)^2 \right| < 1$. So we have,

$$\begin{aligned} \frac{-1/3}{z^2 - 1} + \frac{1/3}{z^2 - 4} &= \frac{(-1/3)(1/z^2)}{1 - (1/z)^2} - \frac{1/12}{1 - (z/2)^2} \\ &= \sum_{n=0}^{\infty} -\frac{1}{3} \frac{1}{z^2} \left(\frac{1}{z} \right)^{2n} - \sum_{n=0}^{\infty} \frac{1}{12} \left(\frac{z}{2} \right)^{2n} \\ &= \sum_{n=0}^{\infty} -\frac{1}{3} \left(\frac{1}{z} \right)^{2n+2} - \sum_{n=0}^{\infty} \frac{1}{12} \frac{1}{2^{2n}} z^{2n} \\ &= \sum_{n=-\infty}^0 -\frac{1}{3} z^{2n-2} - \sum_{n=0}^{\infty} \frac{1}{12} \frac{1}{2^{2n}} z^{2n}. \end{aligned}$$

Region 3: $\{2 < |z| < +\infty\}$

We have $|(1/z)^2| < 1$ and $|(2/z)^2| < 1$.

$$\begin{aligned} \frac{-1/3}{z^2 - 1} + \frac{1/3}{z^2 - 4} &= \frac{(-1/3)(1/z^2)}{1 - (1/z)^2} + \frac{(1/12)(2/z)^2}{1 - (2/z)^2} \\ &= \sum_{n=0}^{\infty} -\frac{1}{3} \frac{1}{z^2} \left(\frac{1}{z} \right)^{2n} + \sum_{n=0}^{\infty} \frac{1}{12} \left(\frac{2}{z} \right)^2 \left(\frac{2}{z} \right)^{2n} \\ &= \sum_{n=0}^{\infty} -\frac{1}{3} \left(\frac{1}{z} \right)^{2n+2} + \sum_{n=0}^{\infty} \frac{2^{2n}}{3} \left(\frac{1}{z} \right)^{2n+2} \\ &= \sum_{n=-\infty}^0 -\frac{1}{3} z^{2n-2} + \sum_{n=-\infty}^0 \frac{1}{3 \cdot 2^{2n}} z^{2n-2} \\ &= \sum_{n=-\infty}^0 \left(-\frac{1}{3} + \frac{1}{3 \cdot 2^{2n}} \right) z^{2n-2} \end{aligned}$$

□

Problem 5. Suppose $f(z)$ analytic on $D = \mathbb{C} \setminus \{0\}$. Show that there is a constant c such that $f(z) - c/z$ has a primitive in D . Give a formula for c in terms of the integral of $f(z)$.

Proof. Note that if $f(z) - c/z$ has a primitive in D , then for any closed curve γ in D we must have $\int_{\gamma} (f(z) - c/z) dz = 0$. Notice that for any closed curve γ' which does not encircle the origin, by Cauchy's Theorem, we can pick any c since both $f(z)$ and c/z are analytic functions and γ' lies in an open connected subset of D , then the integral is 0. So it suffices to find a c that works for closed curves which encircle the origin. Let γ be a closed curve encircling the origin. Define

$$c := \frac{1}{2\pi i} \int_{|z|=1} f(z) dz.$$

Notice that $\int_{|z|=1} \frac{1}{z} dz = 2\pi i$. So we have that

$$\begin{aligned} \int_{|z|=1} f(z) dz - c \int_{|z|=1} \frac{1}{z} dz &= \int_{|z|=1} f(z) dz - 2\pi i c \\ &= \int_{|z|=1} f(z) dz - \frac{2\pi i}{2\pi i} \int_{|z|=1} f(z) dz \\ &= 0. \end{aligned}$$

Now we generalize to arbitrary closed curve encircling the origin. Let γ be a closed curve encircling the origin n times. Then we have

$$\begin{aligned} \int_{\gamma} (f(z) - c/z) dz &= n \int_{|z|=1} (f(z) - c/z) dz \\ &= n \cdot 0 \\ &= 0. \end{aligned}$$

We are done. □

VI.2

Problem 1. Find the isolated singularities of the following functions and determine whether they are removable, essential, or poles. Determine the order of any pole, and find the principle part of each pole.

(a) $z/(z^2 - 1)^2$ (e) $z^2 \sin(1/z)$ (i) $\exp(1/(z^2 + 1))$

Solution for (a): We can factor the denominator to be

$$\frac{1}{(z - 1)^2(z + 1)^2}.$$

So we can see that there are poles of order 2 at ± 1 . We now find the principle part for $z = 1$. We need the Laurent series centered at $z = 1$. Let $w = z - 1$. Then we have

$$\frac{w + 1}{(w^2)(w + 2)^2} = \frac{w + 1}{4w^2} \frac{1}{(1 + w/2)^2}.$$

We can expand $1/(1 + w/2)^2$. We have

$$\begin{aligned} \frac{1}{(1 + w/2)^2} &= \frac{d}{dx} \left(\left(-\frac{2}{1 + w/2} \right) \right) \\ &= \frac{d}{dx} \left(-2 \sum_{n=0}^{\infty} \frac{(-1)^n}{2^n} w^n \right) \end{aligned}$$

$$\begin{aligned}
&= \sum_{n=1}^{\infty} \frac{(-1)^{n+1} (n)}{2^{n-1}} w^{n-1} \\
&= \sum_{n=0}^{\infty} \frac{(-1)^n (n+1)}{2^n} w^n.
\end{aligned}$$

By uniform convergence of compact sets around $w = 0$. So we have,

$$\begin{aligned}
\frac{w+1}{w^2(w+2)^2} &= \frac{w+1}{4w^2} \left(1 - w + \frac{3}{4}w^2 + O(w^3) \right) \\
&= \frac{w+1}{4w^2} - \frac{w+1}{4w} + O(w) \\
&= \frac{1}{4w} + \frac{1}{4w^2} - \frac{1}{4} - \frac{1}{4w} + O(w)
\end{aligned}$$

We only care about terms with negative powers of w . So we have the primitive part is

$$\frac{1}{4w^2} = \frac{1}{4(z-1)^2}.$$

Symmetrically for $w = z + 1$ we end up getting the primitive part is

$$-\frac{1}{4w^2} = -\frac{1}{4(z+1)^2}.$$

□

Solution for (e): Notice that

$$\begin{aligned}
z^2 \sin\left(\frac{1}{z}\right) &= z^2 \left(\frac{1}{z} - \frac{1}{3!} \frac{1}{z^3} + \frac{1}{5!} \frac{1}{z^5} + \dots \right) \\
&= z - \frac{1}{3!} \frac{1}{z} + \frac{1}{5!} \frac{1}{z^3} + \dots
\end{aligned}$$

Clearly there are infinitely many negative powers of z . Hence 0 is an essential singularity. □

Solution for (i): Notice that $1/(z^2 + 1)$ has isolated singularities at $z = \pm i$. We now find the type of singularity for $z = i$ since it is symmetric for the $z = -i$ case. Let $w = z + i$. We have

$$\begin{aligned}
\frac{1}{(z-i)(z+i)} &= \frac{1}{w} \frac{1}{w-2i} \\
&= -\frac{1}{2iw} \frac{1}{1 - (w/2i)} \\
&= -\frac{1}{2iw} \left(1 + \frac{w}{2i} - \frac{w^2}{4} - \frac{w^3}{8i} + O(w^4) \right) \\
&= -\frac{1}{2iw} + \frac{1}{4} + \frac{w}{8i} - \frac{w^2}{16} + O(w^3).
\end{aligned}$$

Now we can plug this into the power series expansion for $\exp$. We have,

$$1 + \left(-\frac{1}{2iw} + \frac{1}{4} + \frac{w}{8i} - \frac{w^2}{16} + O(w^3) \right) + \frac{1}{2} \left(-\frac{1}{2iw} + \frac{1}{4} + \frac{w}{8i} - \frac{w^2}{16} + O(w^3) \right)^2 + \dots$$

Clearly we will have infinite negative powers of w since the first term is a negative power of w and the power grows infinitely. Hence we have infinite negative powers of $z - i$. It follows that i is an essential singularity. Symmetrically $-i$ is an essential singularity.

□